

formal definition of Machine learning:-

Learning:- the ability to improve based on experience.

→ ML explores algorithms that learn from data/build models for data.

→ Model can be used for prediction, decision making or solving.

also called data.

Mitchel definition of ML:-

↳ A computer program is said to learn from Experience (E) with respect to some class of Tasks T and performance measure 'p', if its performance by 'p' improves with Experience

Accuracy.

on tasks 'T' as measured 'E'.

- prediction
- classification
- acting in a environment.

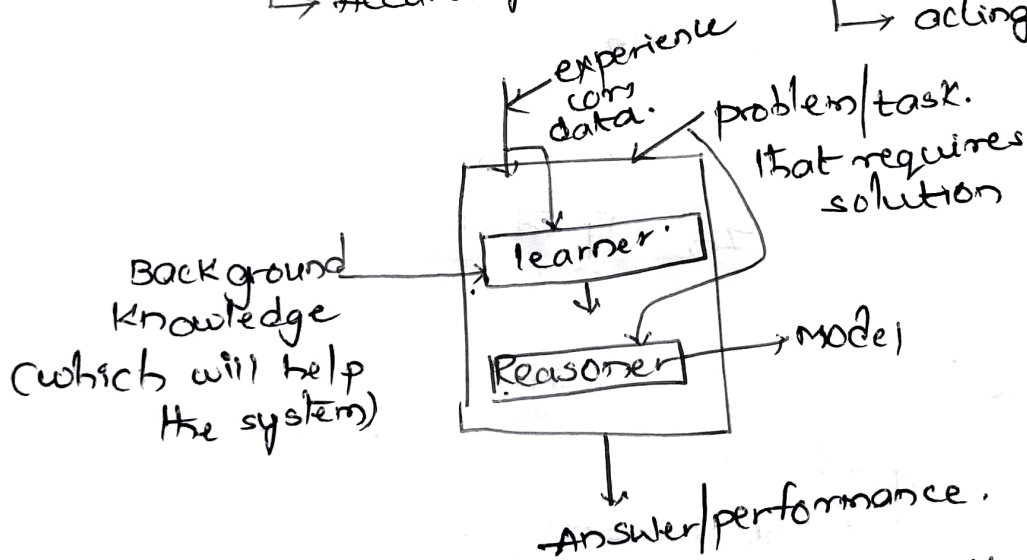


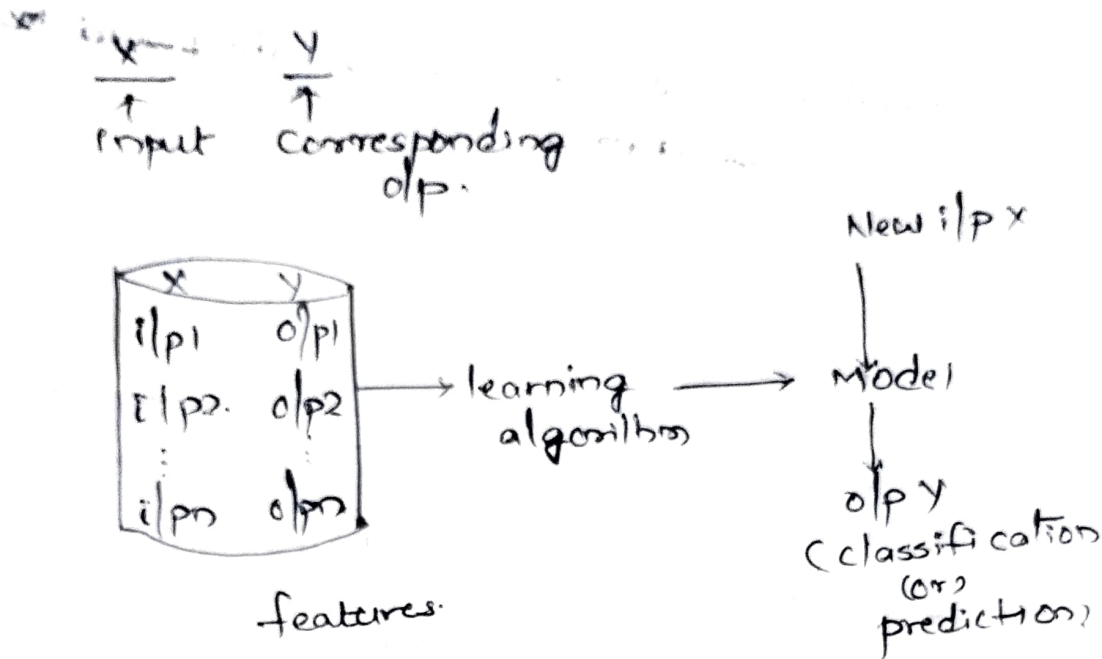
Fig. 1:- schematic diagram of ML model.

For creating a learner:- → features

- (1) choose the training experience (data)
- (2) choose the target-function (that is to be learned)
- (3) choose how to represent the target function
↳ Linear/Non-Linear.

Different types of learning:-

① Supervised learning:-



Example:-

	x_1	x_2	x_3	...	x_n	
Input 1	a_1	a_2	...		a_n	y_1
Input 2	b_1	b_2	...		b_n	y_2
Input 3	c_1	c_2	...		c_n	y_3

The columns $x_1, x_2, x_3, \dots, x_n$ are labeled 'features'. The column y_1, y_2, y_3 is labeled 'target attribute'. The entire dataset is labeled 'training examples'.

For prediction, new input features $z_1, z_2, \dots, z_n$ are fed into the 'predict' box to determine the output.

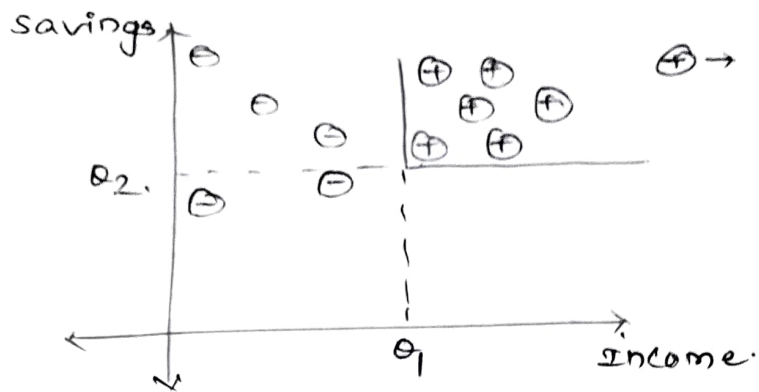
target feature:-

- classification
 - discrete (rain or not rain)
- continuous - regression

ex:-

Location of house	floor area of the house	# of BHKs	<u>price</u>
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Ex: - credit scoring: - (classification).
 x



if $\text{Income} > 0.1$ and $\text{savings} > 0.2$.

Then Low risk.

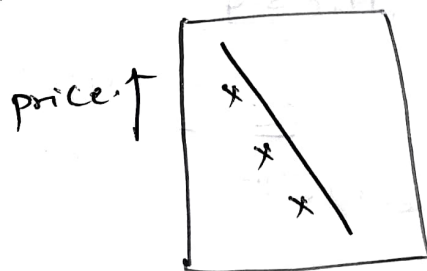
else.

High risk.

classification problem.
 $\uparrow$
 discrete - Low risk
 High risk.

② regression: -

Ex: - price of a used car.



x : - car attributes x : - mileage.

y : - price $\uparrow$ attributes

$$y = g(x; \theta)$$

$\uparrow$
 model

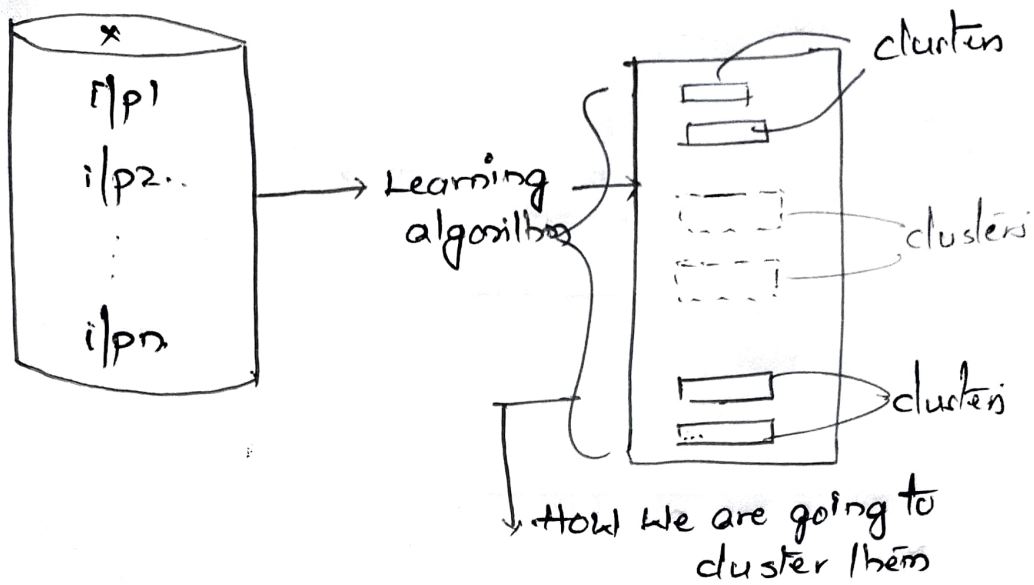
θ : - parameters.

② unsupervised learning: -

x
 $\uparrow$

only x (features)

$\rightarrow$ Given a set of features
 cluster them or summarize them.



Answer:-

based on similarity of data

3. Reinforcement learning:-

↳ type of machine learning, where an agent learns to make decisions by interacting with an environment in order to maximize cumulative reward.

Ex:- self-driving car at a traffic signal

→ Imagine a self-driving car (Agent) approaching a traffic signal (Environment). The car must decide whether to go or stop (Action)

Key components of Reinforcement learning:-

(1) Agent: Learner (or) decision maker

↳ The self driving car making decisions

(2) Environment:- The system the agent interacts with

↳ The traffic system: road, signals and other cars.

(3) states (S):- Current situation of the environment

s_1 : red signal

s_2 : Green signal

(4) Action (A): - set of possible moves taken by the agent.

$$A_1 = \text{stop} ; A_2 = \text{go}.$$

(5) Reward (R): - feedback signal from the environment.

$$R(s_1, A_1) = +10 \text{ (safe behaviour)}$$

$$R(s_1, A_2) = -50 \text{ (traffic violation)}$$

$$R(s_2, A_2) = +20 \text{ (correct action)}$$

$$R(s_2, A_1) = -5 \text{ (unnecessary stop)}$$

(6) policy (π): - ~~overtime~~ strategy used by the agent to select actions.

Overtime, the car learns the following rule (policy)

$$\pi^*(s) = \begin{cases} A_1 (\text{stop}), & \text{if signal} = \text{red} \\ A_2 (\text{go}), & \text{if signal} = \text{green.} \end{cases}$$

(4) Federated learning: -
↳ Is a type of ML model, where multiple clients collaboratively train a model without sharing their private data.

Key components of Federated learning: -

Component:

(1) clients

(2) Local model

(3) server (co-ordinator)

(4) communication

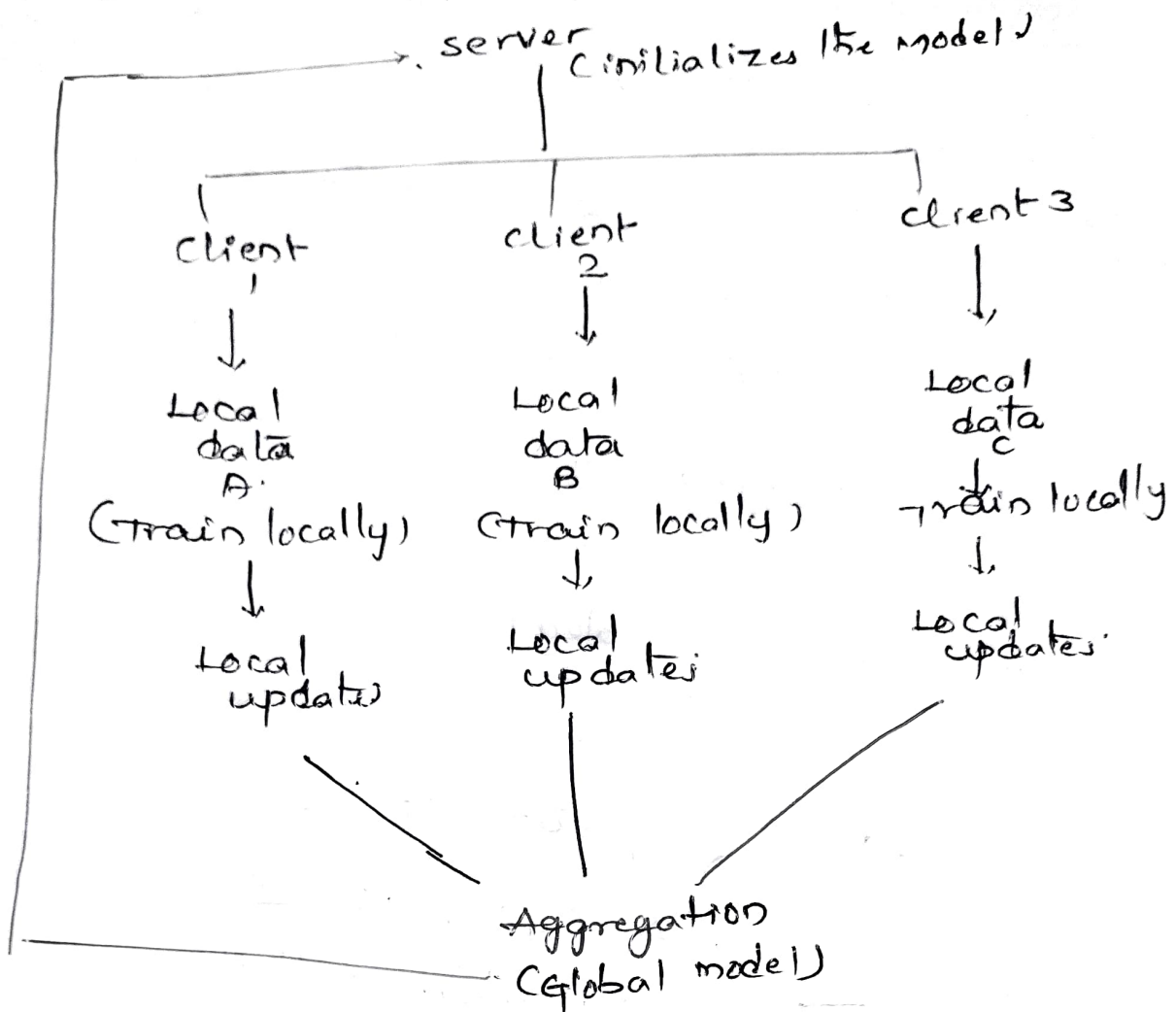
Role: -

Devices (e.g.: phones, hospital)
that holds local
private data.

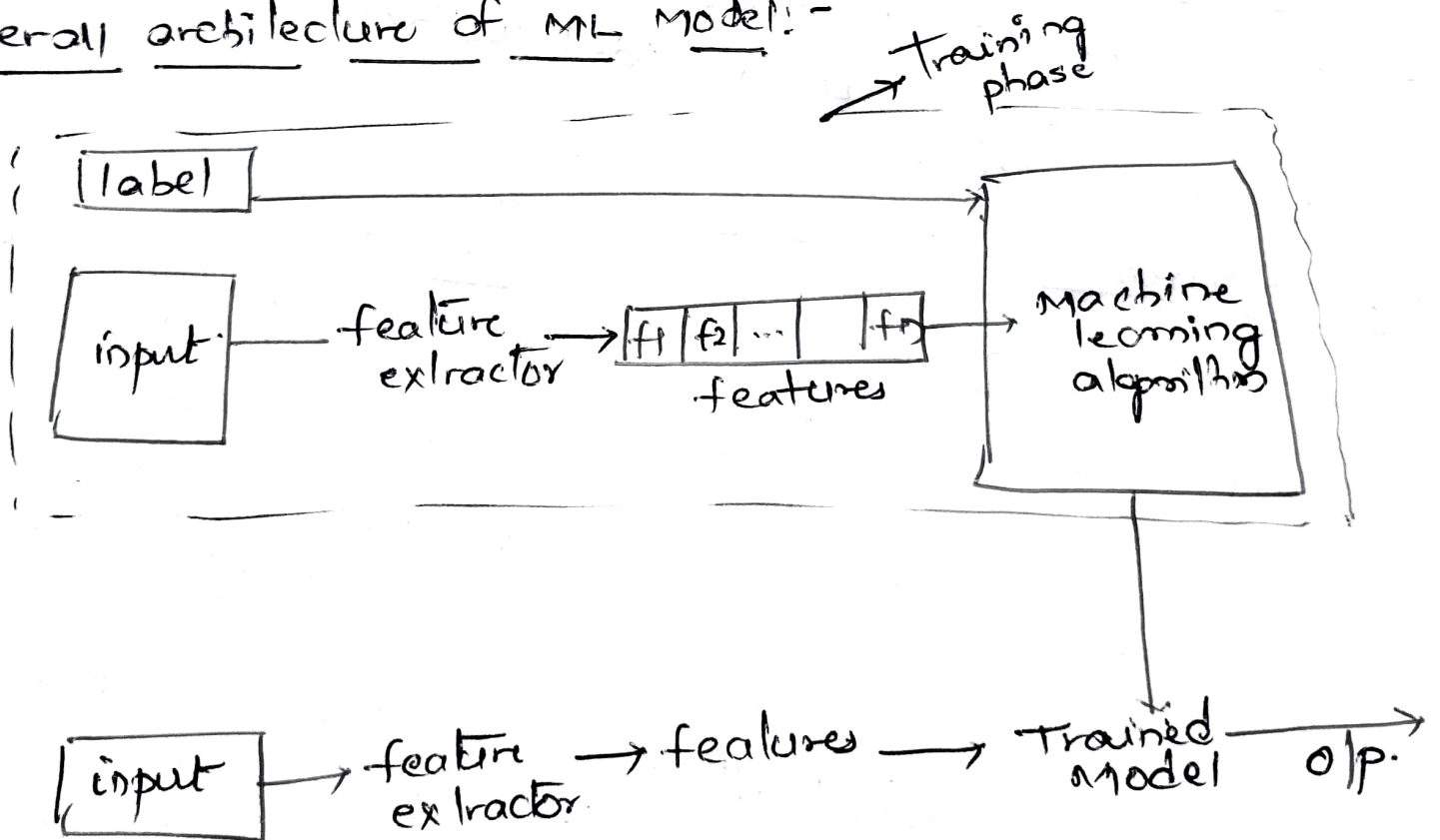
A copy of the model that
is trained using
local data.

combines all updates
from clients and
aggregates the model.

sharing of model weights
between server and
clients.



→ overall architecture of ML Model:-



Given examples target (or) output attribute:

$(\hat{x}, y)$
 $\uparrow$
 values of
 different
 features

$(\hat{x}, f(\hat{x}))$
 $\uparrow$
 features

function we are
 going to learn on
 i/p features

for classification

$f(\hat{x})$ is discrete $\Rightarrow y = f(\hat{x})$

for regression

$f(\hat{x})$ is continuous $\Rightarrow y = f(\hat{x})$

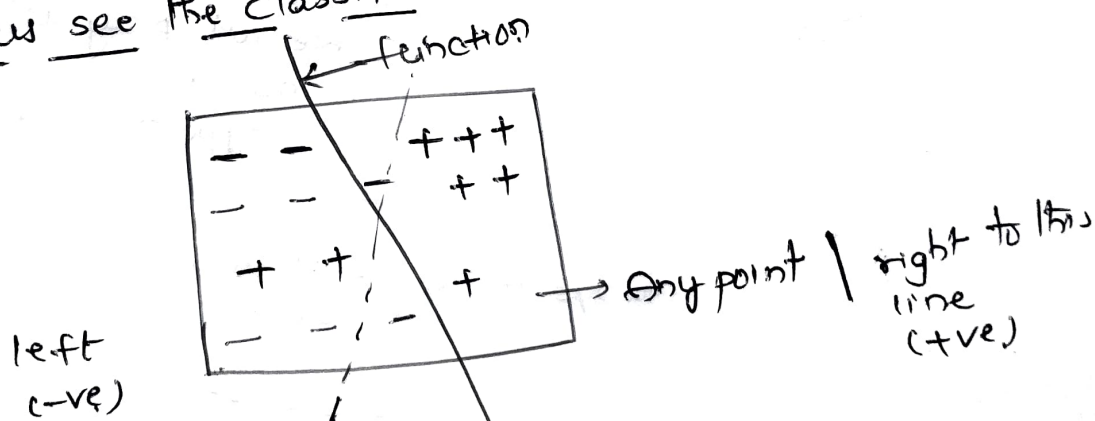
for probability estimation $\Rightarrow f(\hat{x}) = \text{probability of } \hat{x} \Rightarrow y$

These are called inductive learning problems.

Why inductive learning

Given some data, we are trying to do some induction to try to identify a function which can explain the data.

Ex! - Let us see the classification problem! -



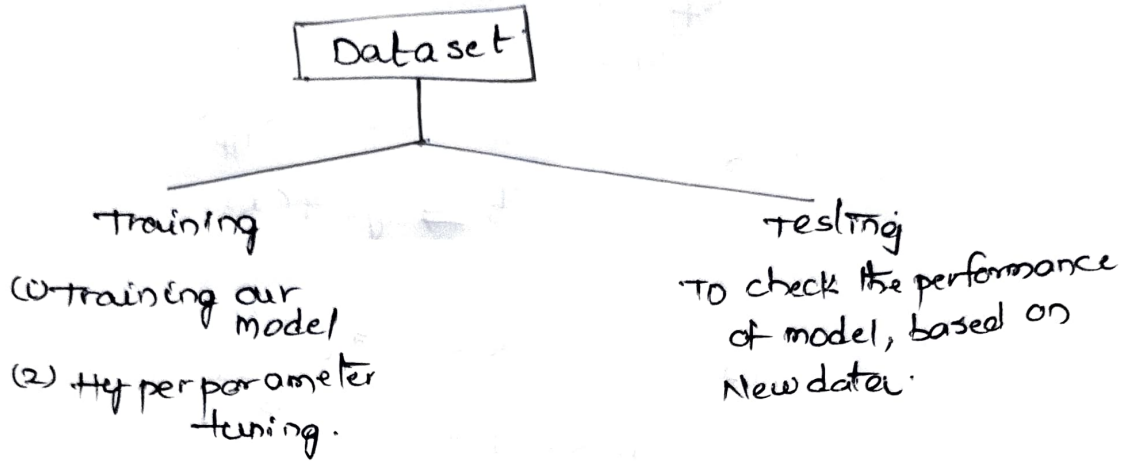
Hypothesis space! - set of legal hypothesis

hett

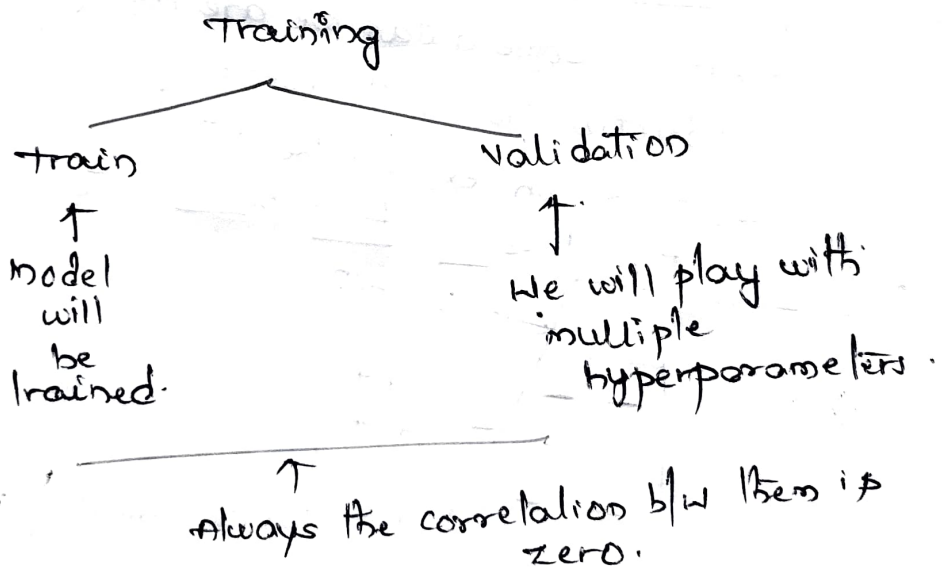
$\uparrow$ Hypothesis space

o/p of a learning algorithm that better fit the data

$h \in H$
 $\uparrow$
 find 'h' that
 better defines the
 data.



Always:-
 correlation b/w the two sets will
 be zero.



validation:-

Let us take simple example,

	X	Y
training	1	2
	2	4
validation	3	6
testing	4	8

Let us do training now! -

(1) Degree 1 (or) Linear regression.

$$y = a_1 x + a_0$$

$$a_0 = 0$$

$$\rightarrow y = 2x$$

↑
training data.

prediction on validation data! -

$$x = 3 \Rightarrow y = 6$$

$$\text{validation error: } - (6-6)^2 = 0.$$

(2) Degree 2! - $y = a_2 x^2 + a_1 x + a_0$
Let us assume a simple model

$$y = x^2 - x + 2$$

$$x = 3 \Rightarrow y = 8$$

$$\text{error: } - (8-6)^2 = 4$$

Degree 3! -

$$y = x^3 - 2x^2 + 3x$$

$$x = 3 \Rightarrow y = 18$$

$$\text{error: } - (18-6)^2 = 144$$

<u>Degree</u>	<u>prediction</u>	<u>Error</u>
	6	0
1		4
	8	
2		144
	18	
3		

$\Rightarrow$ Error is Low for degree 1. $\therefore$ We choose a simple model with degree 1 as the hyperparameter

for test data

$$y = 2x; x = 4$$

$$\hat{y} = 8$$

$$(8-8)^2 = 0$$

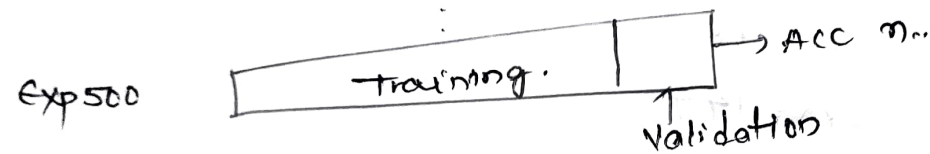
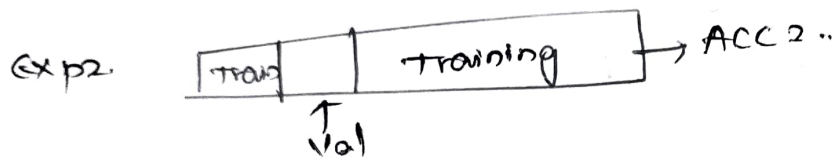
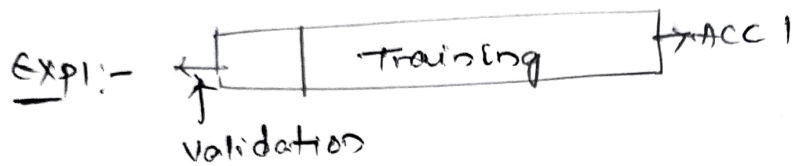
(check validation - theory pdf for better understanding of validation) and drawbacks.

disadvantages of Validation

- Validation approach is sensitive to specific split used.
- cross validation reduces variance by testing across multiple splits.
- it ensures all samples are used in validation atleast once.

Types of cross-validation:-

(i) Leave one out CV (LOOCV)



disadvantage:-

- (1) only one record as validation
- (2) Time complexity of Training model increases.

Overfitting:- Training Acc $\uparrow\uparrow$
Validation " $\downarrow\downarrow$

New test data: $\rightarrow$ Acc $\downarrow\downarrow$.

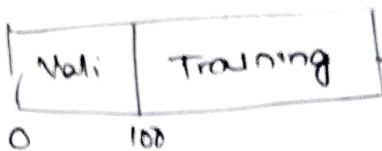
② Leave p out CV

$p=10; p=20; p=30$

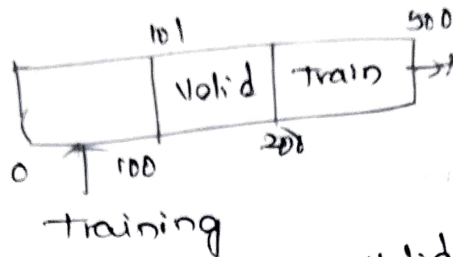
③ K-fold cross validation:-

$$n=500; k=5$$

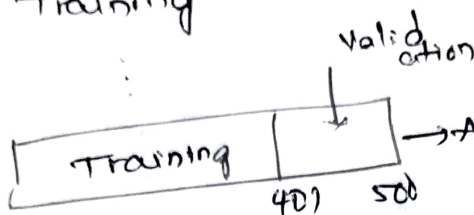
Exp 1:-



Exp 2:-



Exp 5:-



total
dataset size

$$\text{Validation size} = \frac{500}{5} = 100$$

Accuracy
= Average
Accuracy.

problem:-

→ classification problem, part by part.

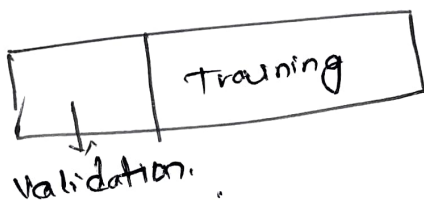
→ one set of class may come for validation

→ testing this might be an issue

④ stratified K-fold validation

$$k=5; n=500$$

$$= \frac{500}{5} = 100$$



100:-

60:- 40

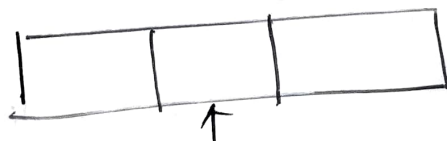
60

50:- 50

11

60 → 1's

40 → 0's



proportion
with respect
to
outcomes

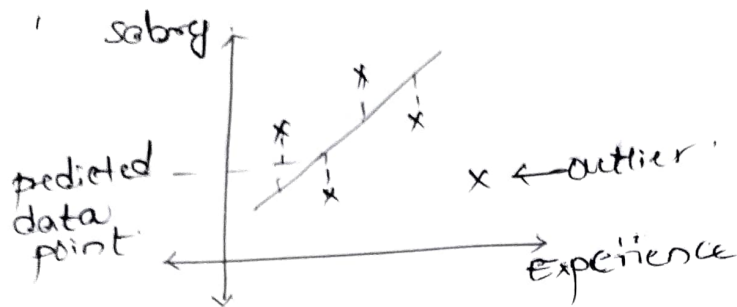
solve one problem

pdf

more details (document)
cross-validation-theory

Performance metrics:-

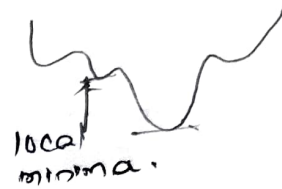
Regression performance metrics:-



(1) MSE (Mean squared error)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

↑
Quadratic equation.



Advantages:-

- (1) Differentiable
- (2) It has one local (or) global minima.
- (3) Converges faster.

Disadvantage:-

- (1) Not robust to outliers
↑
error will increase.

$$(y_i - \hat{y}_i)^2$$

↑
y_i, y_i
↑ high

↑
error will not be in same units.

(2) Mean absolute error:-

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

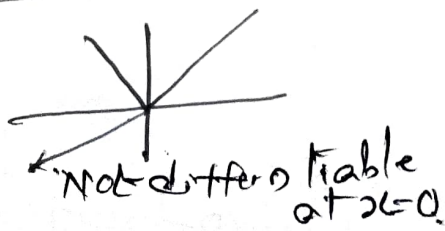
Advantage:-

- (1) Robust to outlier
MAE ↑↑, not like MSE.

(2) It will be in the same unit.

(2) Convergence takes more time
optimization is a complex task.

Disadvantages:-



3. Root mean squared error:-

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

Advantage:-

(1) The values are in some unit.

(2) Differentiable.

(4) Pearson linear correlation coefficient (PLCC)

$$PLCC = \frac{COV(X, Y)}{\sigma_X \sigma_Y}$$

(5) Spearman Rank order Correlation coefficient.

$$SRCC = \frac{1 - 6 \sum d^2}{n(n^2 - 1)}$$

d = difference between ranks.

⑥ R-squared error:-

is given by formula:

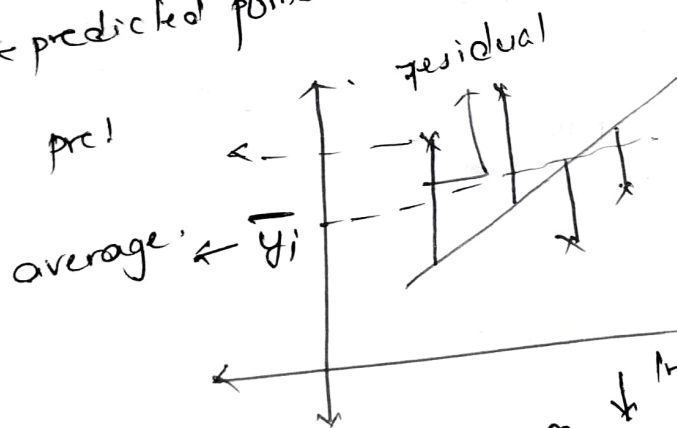
$$= 1 - \frac{SS_{Res}}{SS_{Total}}$$

sum of squared residual

y_i ← true points

$\hat{y}$ ← predicted points

sum of squared total



$$\Rightarrow 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2}$$

average of true points

True point

if best fit line passes through True points $(y_i - \hat{y}_i)^2 \Rightarrow$ smaller

$\Rightarrow 1 - \frac{\text{small number}}{\text{large number}}$

$= 1 - \text{smaller number}$

≈ 1

(0.70 $\rightarrow$ 70% accurate)

0.85 $\rightarrow$ 85% accurate

$\Rightarrow$ more this value, more accurate is my model.

⑦ adjusted R-squared error:-

dataset:-

size of the
House
 $\downarrow$
increases

price of the
House
 $\downarrow$
price increases $\Rightarrow$ +ve correlation
 $\uparrow$
positive.

Let us compute

R-squared error = 75% = 0.75

add one more feature to dataset

No. of bed
rooms
Location.

price $\Rightarrow$ +ve correlation
increases

R-squared error = 85% = 0.85

Add $\downarrow$

Gender $\rightarrow$ NO correlation.

R-squared error = 87% = 0.87

$\uparrow$
More number of features
there is a problem

Students
self-learning
 $\downarrow$
do in lab

$\Rightarrow$ problem of R-squared error.

To prevent this:-

$$\text{Adjusted } R^2\text{-error} = 1 - \frac{(1 - R^2)(N-1)}{N-p-1}$$

$N \leftarrow$ No. of data points

$p \leftarrow$ No. of independent features

if $p = 2$; $R^2 = 90\% \Rightarrow R^2\text{-adjusted} = 86\%$

$p = 3$; $R^2 = 92\% \Rightarrow R^2 - " = 82\%$

classification performance metrics

Dataset: -

	o/p	predicted
	1	0
1	0	1
1	1	1
1	0	0
1	1	1
1	1	1
0	0	1
0	1	0

Confusion matrix → gives details about the classification and misclassification between Actual and predicted values.

predicted value

	1	0
1	3	2
0	1	1

Actual

diagonal element = correct predictions

Notation: -

predicted

	1	0
1	TP	FP
0	FN	TN

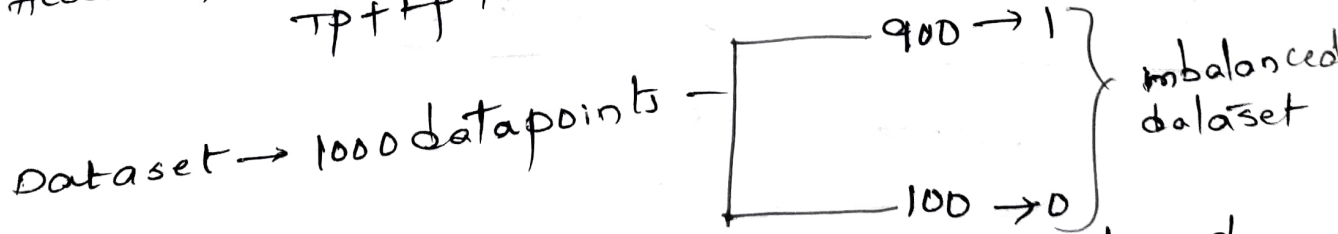
Actual

Actual is 0, predicted 1

TP, TN are correct predictions

performance metrics: -

(i) Accuracy = $\frac{TP + TN}{TP + FP + FN + TN} = \frac{3 + 1}{3 + 2 + 1 + 1} = \frac{4}{7}$



ratio of o/p categories = 9:1 ⇒ Not good.

imbalanced dataset) accuracy → Not good.
90% accuracy ← 1 ← most test cases.

② precision:-

↳ out of all the Actual values, how many are correctly predicted

$$\text{prec} = \frac{TP}{TP + FP}$$

Actual

TP	FP
FN	TN

predi

precision

③ Recall:-

↳ out of all the predicted values, how many are correctly predicted.

$$\text{Recall} = \frac{TP}{TP + FN}$$

(CTPR)

Actual

TP	FP
FN	TN

Recall

uses cases:-

(i) spam classification:-

		Actual		
		spam	Not spam	
pred	spam	TP	FP	↓ ↓ ↓ ← We should reduce
	Not spam	FN	TN	
		Mail → spam Model → spam		} Good

Mail → Not spam } blunder
Model → spam

$$\text{precision} = \frac{TP}{TP + FP}$$

↑
Most priority
will be given to FP

② TO predict whether _____ person has diabetes or Not:-

Truth → diabetes
Model → doesn't have diabetes } blunder

Truth → diabetes } good
Model → "

Truth $\rightarrow$ Not having diabetes } not blunder $\Rightarrow$ 2nd opinion
 model $\rightarrow$ diabetes } $\Downarrow$ (Hospital) check

Diabetes	TP	FP
No diabetes	FN	TN

We should reduce

use case of diseases

$$\text{Recall} = \frac{TP}{TP + FN}$$

④ F-beta score:-

$$F_\beta = (1 + \beta^2) \frac{\text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}}$$

if Fp and FN are both important $\Rightarrow \beta = 1$

$$(1) F_1 \text{ score} = (1 + 1^2) \frac{\text{precision} \times \text{Recall}}{\text{precision} + \text{Recall}}$$

$$\Rightarrow 2 \times \frac{\text{prec} \times \text{Rec}}{\text{prec} + \text{Rec}}$$

(2) Fp is more important FN; $\beta = 0.5$

$$F_{0.5} = 1.25 \times \frac{P \times R}{P + R}$$

if $FN \gg FP$; $\beta = 2$.

$$F_2 \text{ score} = 5 \times \frac{P \times R}{P + R}$$

⑤ Matthews correlation coefficient

$\rightarrow$ takes into account TP, TN, FP and FNs and is generally regarded as a balanced measure even when the classes are imbalanced.

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$MCC = 1 \Rightarrow$ perfect prediction

$= -1 \Rightarrow$ total disagreement

multiclass:-

Actual

FP
Actual

predicted

	A	B	C
A	$TP_A = M_{AA}$	$FP_{B \rightarrow A} = M_{AB}$	$FP_{C \rightarrow A} = M_{AC}$
B	$FP_{A \rightarrow B} = M_{BA}$	$TP_B = M_{BB}$	$FP_{C \rightarrow B} = M_{BC}$
C	$FP_{C \rightarrow A} = M_{CA}$	$FP_{B \rightarrow C} = M_{CB}$	$TP_C = M_{CC}$

$$\text{Accuracy} = \frac{M_{AA} + M_{BB} + M_{CC}}{\sum_i \sum_j M_{ij}} = \frac{TP_A + TP_B + TP_C}{N}$$

precision:-

$$\text{precision}_A = \frac{M_{AA}}{M_{AA} + M_{AB} + M_{AC}} = \frac{TP_A}{\text{predicted as A}}$$

$$\text{precision}_B = \frac{M_{BB}}{M_{BB} + M_{BA} + M_{BC}} = \frac{TP_B}{\text{predicted as B}}$$

$$\text{precision}_C = \frac{M_{CC}}{M_{CC} + M_{CA} + M_{CB}} = \frac{TP_C}{\text{predicted as C}}$$

Macro-averaged precision (mean precision)

$$\text{Mean precision} = \frac{1}{3} (\text{precision}_A + \text{precision}_B + \text{precision}_C)$$

Recall per

$$\text{Recall}_A = \frac{M_{AA}}{M_{AA} + M_{BA} + M_{CA}} = \frac{TP_A}{\text{Actual as A}}$$

$$\text{Recall}_B = \frac{M_{BB}}{M_{BB} + M_{AB} + M_{CB}} = \frac{TP_B}{\text{Actual as B}}$$

$$\text{Recall}_C = \frac{M_{CC}}{M_{CC} + M_{AC} + M_{BC}} = \frac{TP_C}{\text{Actual as C}}$$

Macro-average Recall (Mean Recall)
 $\Rightarrow \frac{1}{3}(\text{Recall}_A + \text{Recall}_B + \text{Recall}_C)$

F1 score:-

Mean F1 score $\neq 2 \cdot \frac{\text{Mean precision} \times \text{Mean recall}}{\text{Mean precision} + \text{Mean recall}}$

$$F1_A = \frac{2 \times \text{precision}_A \times \text{Recall}_A}{\text{precision}_A + \text{Recall}_A}$$

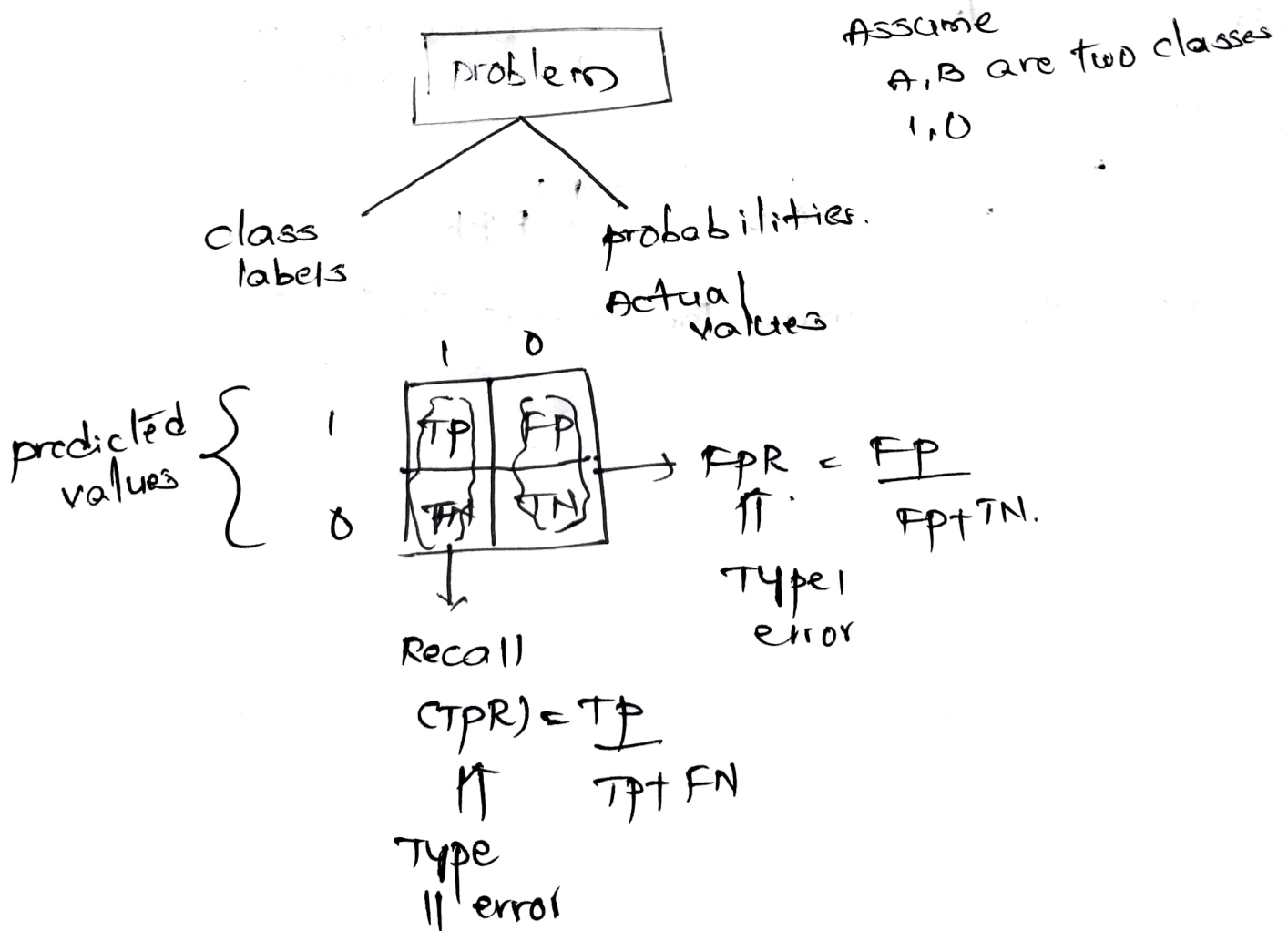
$$F1_B = \dots$$

$$F1_C = \dots$$

Macro Averaged (F1-score) (or) Mean F1-score
 $= \frac{1}{3}(F1_A + F1_B + F1_C)$

How MCC $\leftarrow$ self learning.

Roc and AUC



y	$\hat{y}$	$\hat{y}(0)$	$\hat{y}(0.2)$	threshold values
1	0.8	1	1	$\Rightarrow [0, 0.2, 0.4, 0.6, 0.8]$
0	0.96	1	1	> 0.2
1	0.4	1	1	
1	0.3	1	1	
0	0.2	1	0	
1	0.7	1	1	

$$\hat{y}(0) = \frac{4}{4+0} = \text{TPR} = 1$$

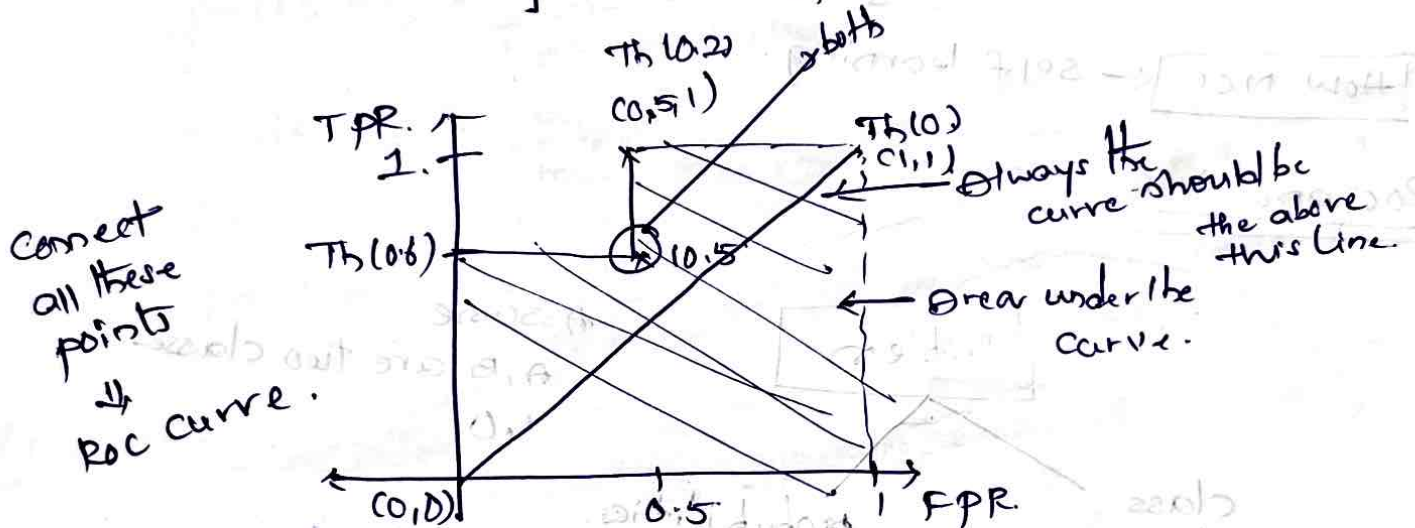
$$\text{FPR} = \frac{2}{2+0} = 1$$

$$\hat{y}(1) = 1$$

$$\text{TPR} = 1$$

$$\text{FPR} = \frac{1}{1+1} = \frac{1}{2} = 0.5$$

focus



$\Rightarrow$ More the area under the curve, better the model

dataset! - (data scientist point of view? (domain expert?))

* What threshold point need to be selected?

① higher TPR. Th 0.62
TPR; FPR = 0

② higher TPR, don't care about false positive rate

Th(0.2)
 $\rightarrow$ use this threshold.

③ I've want to focus both

Naive Bayes Theorem

→ Naive Bayes Algorithm:-

→ used for both binary and multiclass classification problem.

Independent events:- Rolling a dice = {1, 2, 3, 4, 5, 6}

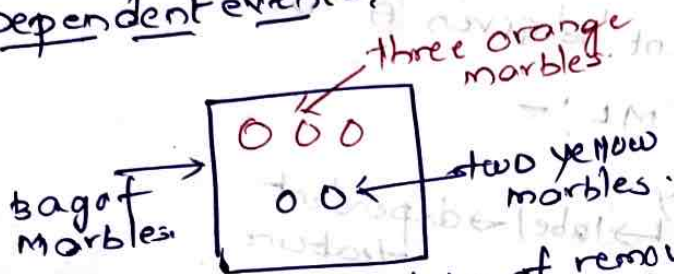
↑
specific outcomes.

$$pr(1) = \frac{1}{6}; pr(2) = \frac{1}{6}; pr(3) = \frac{1}{6}$$

$$pr(4) = \frac{1}{6}; pr(5) = \frac{1}{6}; pr(6) = \frac{1}{6}$$

one outcome is not changing the outcome of other.

* Dependent events:-



(c) what is the probability of removing orange marble and then yellow marble?

$$p(o) = \frac{3}{5} \rightarrow 1^{st} \text{ event} \rightarrow \text{orange marble has been removed.}$$

A rectangular box represents a bag of marbles after one orange marble has been removed. Inside the box, there are two red circles (orange marbles) and two black circles (yellow marbles). An arrow points from the text 'probability of 1st event is impacting the second event' to the box. Another arrow points from the text 'given orange event has taken place' to the box.

$$p(y) = \frac{2}{4} = \frac{1}{2} \rightarrow \text{probability of 1st event is impacting the second event.}$$

$$p(y|o) \leftarrow \text{given orange event has taken place.}$$

$$p(o \text{ and } y) = p(o) * p(y|o)$$

↓ ↓
1st event 2nd event Conditional probability

$$\Rightarrow \frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$$

Generic way:-

$$P(A \text{ and } B) = P(A) * P(B|A)$$

Bayes Theorem:-

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$\Rightarrow P(A) * P\left(\frac{B}{A}\right) = P(B) * P\left(\frac{A}{B}\right)$$

$$\boxed{P(A|B) = \frac{P(A) * P\left(\frac{B}{A}\right)}{P(B)}} \rightarrow \text{Bayes Theorem.}$$

$\Rightarrow P(A|B)$ = probability of Event 'A' given B has occurred.

$P(A)$ = probability of event 'A'

$P(B)$ = " " " 'B'

$P(B|A)$ = probability of event 'B' given 'A' has occurred.

How Bayes Theorem is used in ML:-

$x_1, x_2, x_3, \dots$ $\rightarrow$ independent features
 Y $\rightarrow$ label $\rightarrow$ dependent feature

— — — — — yes

— — — — — NO

— — — — — yes

— — — — — NO

$$P(Y|x_1, x_2, x_3) = \frac{P(Y) * P(x_1, x_2, x_3|Y)}{P(x_1, x_2, x_3)}$$

$$\Rightarrow \frac{P(Y) * P(x_1|Y) * P(x_2|Y) * P(x_3|Y)}{P(x_1) * P(x_2) * P(x_3)}$$

regarding the dataset:-

$$P(\text{yes}|x_1, x_2, x_3) = \frac{P(\text{yes}) * P(x_1|\text{yes}) * P(x_2|\text{yes}) * P(x_3|\text{yes})}{P(x_1) * P(x_2) * P(x_3)}$$

$$pr(NO|x_1, x_2, x_3) = \frac{pr(NO) * pr(x_1|NO) * pr(x_2|NO) * pr(x_3|NO)}{pr(x_1) * pr(x_2) * pr(x_3)}$$

Denominator of both $pr(yes|x_1, x_2, x_3)$ and $pr(NO|x_1, x_2, x_3) \rightarrow \text{constant}$.

problem! — (dataset, check LMS)

outlook

	yes	NO	$P(E yes)$	$P(E NO)$
sunny	2	3	2/9	3/5
overcast	4	0	4/9	0/5 = 0
Rain	3	2	3/9	2/5
	<u>9</u>	<u>5</u>		

Temperature! —

	yes	NO	$P(E yes)$	$P(E NO)$
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
cool	3	1	3/9	1/5

$$yes = 9 ; NO = 5$$

$$P(y) = 9/14 ; P(N) = 5/14$$

New test data:

Test (sunny, hot)

$$pr(yes|sunny, hot) = pr(yes) * pr(sunny|yes) * pr(hot|yes)$$

$$\rightarrow \frac{9}{14} * \frac{2}{9} * \frac{2}{9}$$

$$= \frac{2}{14} = 0.071$$

① multinomial Naïve Bayes:-

Input data in the form of text.

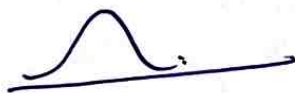
Dataset:- spam classification.

	<u>Email body</u>	<u>spam/Not-spam</u>
text {	you have won million dollar lottery	spam.
	Hi, you have done good job !!	Not-spam.

Numerical $\Rightarrow$ NLP techniques
values

② Gaussian NB:-

if the features are following a gaussian distribution.
then we use gaussian NB.



iris dataset:-

Age Height weight yes/no.

continuous
features {

$\Rightarrow$ If some of features use Bernouli and gaussian distribution
 $\Rightarrow$ max (Bernouli, gaussian).

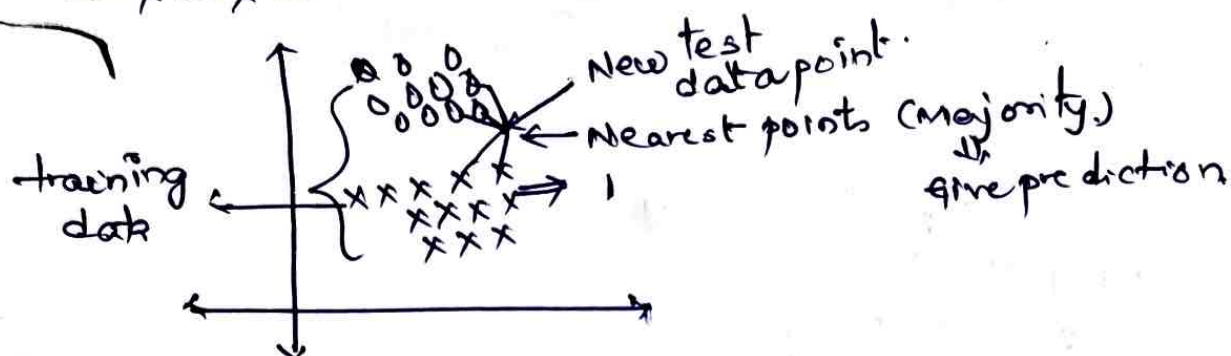
K-nearest Neighbor (KNN)

→ can be used to solve

(i) classification

(ii) Regression

(i) classification:-



Steps:-

(1) We have to initialize the 'k' value.

$k > 0, 1, 2, \dots, \infty$

i.e. $k = 1, 2, 3, \dots$

How to find 'k' value $\Rightarrow$ Hyperparameter tuning.

(2) find the 'k' nearest Neighbor for the test data.

Let us assume

$k = 5$

Top '5' Nearest

(3) from those $k=5$ values, how many neighbors belong to '0' category and '1' category.

$0 \rightarrow 1$
 $1 \rightarrow 4$ } 0/5 is predicted as '1'

Euclidean distance:-

(x_1, y_1) (x_2, y_2)

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

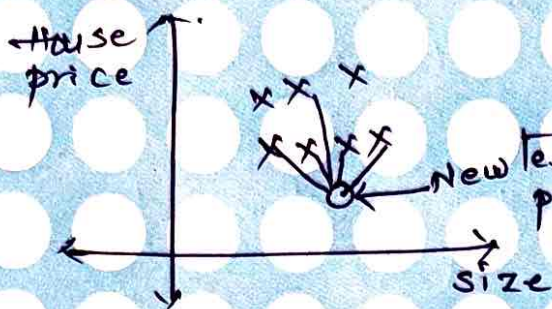
Manhattan distance.

(x_1, y_1) (x_2, y_2)

$$\text{Neighbors distance} = |x_1 - x_2| + |y_1 - y_2|$$

Regression:-

Elphydin



New test point = Average of all these points to find out the o/p if we have outlier $\downarrow$ median.

Example:-

training data:-

point	value x	class
A	1	Red.
B	2	Red.
C	3	Red
D	8	blue
E	9	blue
F	10	blue.

Red class near small values.
Blue class near large values.

Test point:-

$x_{test} = 4$

Case (1)

$K=1$

Nearest Neighbor to 4:- This point can be an outlier

Neighbor	Distance	class
(3)	1	Red.

Case (2)

$K=3$

Neighbor
3
2
1

Distance
1
$\frac{2}{3}$

class
Red $\Rightarrow$ Majority class = Red.
Red.
Red

Case (3):-

$K=5$

Neighbor	class
3	Red
2	Red.
1	Red
8	blue.
9	blue.

majority class:- Red.
decision:- Red.
still reasonable

$K=6$

(1) All training samples

class	Count
Red	3
blue	3

(Tie)

Case (5)

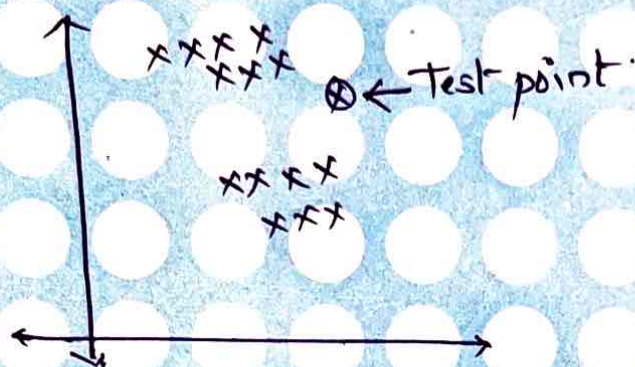
$k \rightarrow \infty$
 $\rightarrow$ model ; KNN ignores distance completely

predicted class = global majority class

Model becomes distance independent.

$k = \text{small} \Rightarrow$ highly local, Noisy

Variants of K-NN:-



(1) Compute the Nearest Neighbor
(1) find the distance between all the points and then take the top 5 Neighbors

(2) Think about time complexity
 $O(N)$

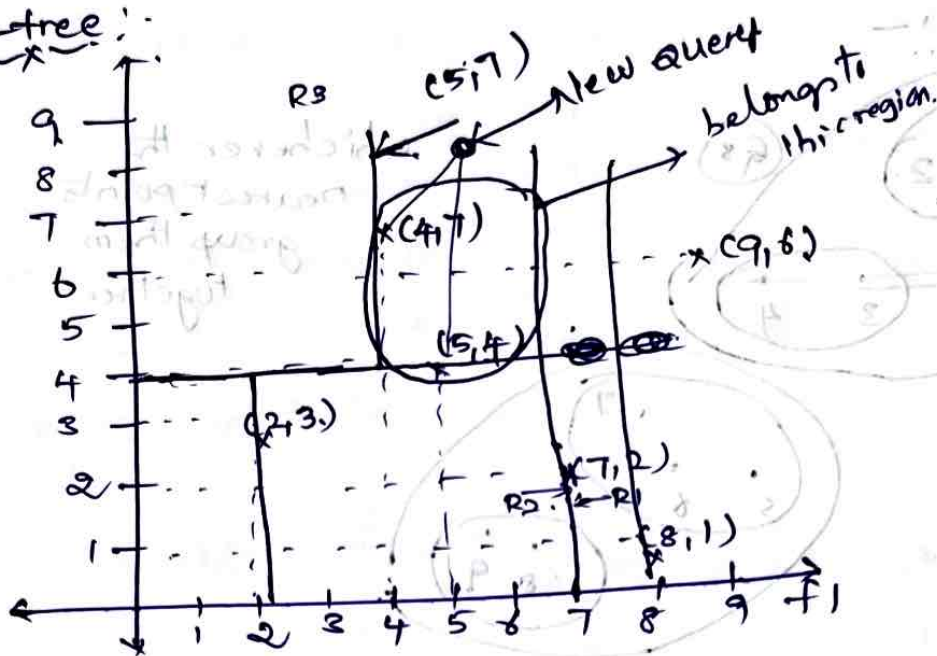
$\uparrow$
millions of dataset

How can we decrease the time complexity:-

Let us assume, we have the dataset.

f1	f2
7	2
5	4
9	6
2	3
4	7
8	1

(1) KD-tree:



f1:- 2, 4, 5, 7, 8, 9
Median = 5
f2:- 1, 2, 3, 4, 6, 7
 $\frac{3+4}{2} = \frac{7}{2} = 3.5$

Consider next point = 7) project along f1-axis

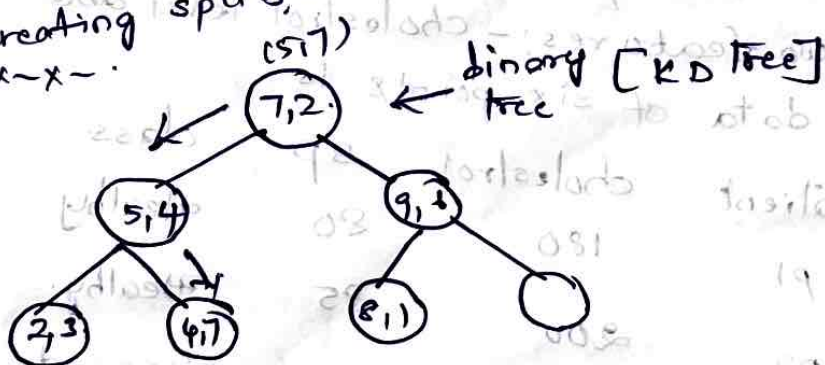
Step 2:-

(9,6), (8,1)

9, 8

$$\Rightarrow \frac{9+8}{2} = \frac{17}{2} = 8.5$$

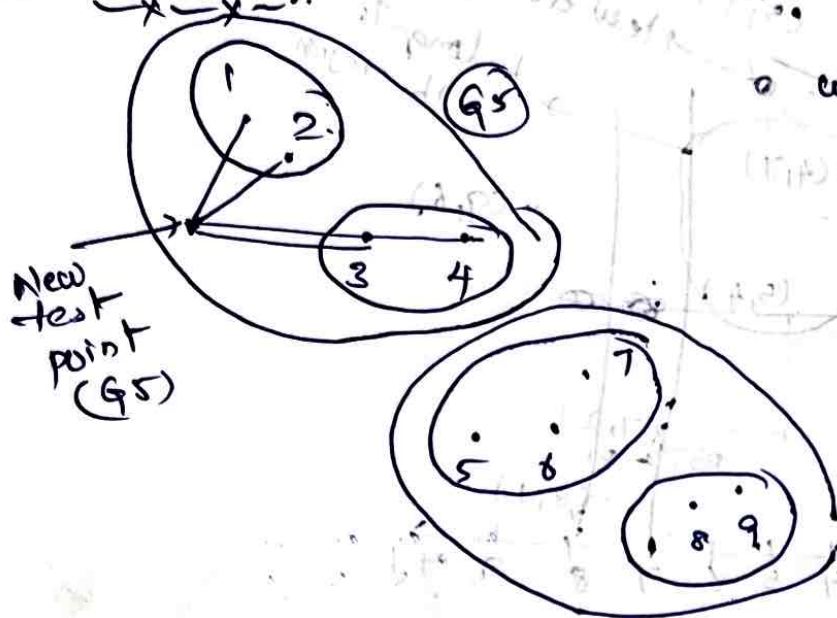
We are creating splits:-



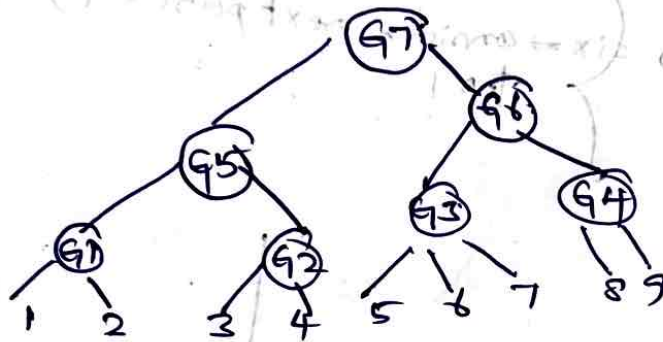
Time complexity is reduced.

Back tracking

② Ball-tree! -



whichever the nearest points group them together



problem! -

(1) A Hospital wants to use K-Nearest Neighbor (KNN) algorithm to predict whether a patient is healthy (or) diseased based on two features! - cholesterol level and blood pressure.

The training data of six points is

patient	cholesterol	BP	class
p1	180	80	Healthy
p2	200	85	Healthy
p3	220	90	Diseased
p4	240	160	Healthy
p5	190	70	Healthy
p6	230	95	Diseased

Using KNN with $k=3$, compute the nearest Neighbors of each new point (using euclidean distance), perform majority voting to classify each as healthy (or) diseased, and explain the choice of 'k'.

plain the choice of 'K'

parent (210, 88)

Euclidean distance: $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

square root does not have impact

patient: d^2 to A.

clam

P1 $(210-180)^2 + (80-80)^2 = 964$ healthy

Healthy.

p2 $(10)^2 + (3) = 109$

Healthy.

93 104

Diseased

pg 1044

Diseased

p5 724

Healthy

pb 449

Diseased

Three smallest distances:-

test distances:-
 $p_3(104, D)$, $p_2(109, H)$, $p_6(1449, D)$

 D, H, D

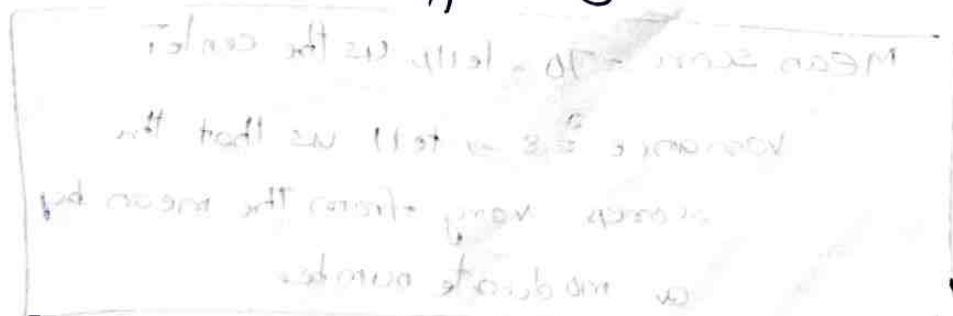
majority = diseased.

with $k=1 \Rightarrow$ prediction depends on single closest
and can be very sensitive to noise (an outlier)

$k=3$ (moderate choice)

$k = \infty \Rightarrow$ distance independent

o/p: - majority Global class



Mean - Mean (or) average is a measure of the center of a set of numbers.

~~refer overleaf~~

Ex: - Suppose a student takes 5 tests and gets the following scores

70, 72, 68, 74, 66

$$\text{Mean} = \frac{70 + 72 + 68 + 74 + 66}{5}$$

$$\text{Mean} = 70$$

(or)

Average.

It gives us a simple way to summarize all the scores within the number.

Variance -

↳ Variance tells us how spread out the numbers are from the mean. It shows how much the values vary.

Ex: - some test scores: -

-x-

70, 72, 68, 74, 66

$$\text{Mean} = 70$$

$$\text{Variance} = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2$$

$$\Rightarrow \frac{0 + 4 + 4 + 16 + 16}{5}$$

$$= 8$$

Mean score = 70 = tells us the center

Variance = 8 = tells us that the

scores vary from the mean by a moderate number.

- * if all scores were exactly the same, variance would be 0.
- if scores were very different, variance would be high.

Bias: -

-x-x-

→ tell us how far the average model prediction is from the actual (true) value.

→ high bias means the model systematically makes wrong predictions.

$$\text{Bias} = E[\hat{f}(x)] - f(x)$$

problem context: -

→ we aim to predict a student's exam score based on the number of hours studied.

Let us assume that the true underlying function is

$$f(x) = 5x + 50$$

x = No. of hours studied

$f(x)$ = exam score.

Let us assume; if a student studies for '4' hours

$$f(4) = 5 \times 4 + 50 = 70$$

Case (i): - Low bias, Low variance (ideal case): -

-x-x-

Let us assume if we have three different models.

$$\text{Model 1: } \hat{f}_1(x) = 5.0x + 49.9$$

$$2: \hat{f}_2(x) = 4.9x + 50.1$$

$$3: \hat{f}_3(x) = 5.1x + 50.0$$

predictions at $x=4$

$$\hat{f}_1(4) = 49.9; \hat{f}_2(4) = 69.7; \hat{f}_3(4) = 70.4$$

$$E[\hat{f}(4)] = \frac{49.9 + 69.7 + 70.4}{3}$$

3

$$= 70.0$$

$$\text{Bias} = E[\hat{f}(4)] - f(4)$$

$$\rightarrow 70 - 70 = 0$$

$$\text{Variance} = \frac{1}{3} [(69.9 - 70)^2 + (69.7 - 70)^2 + (70.4 - 70)^2]$$

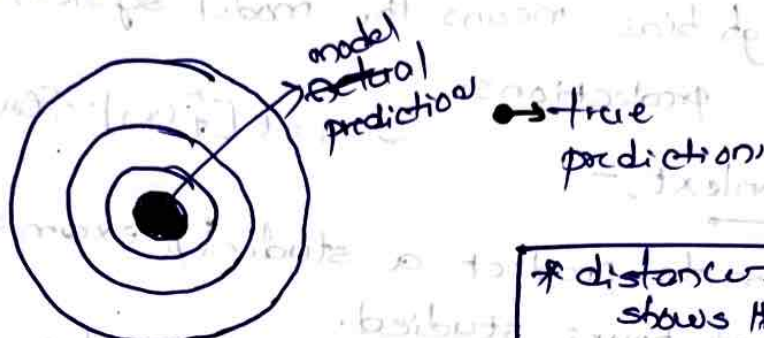
$$\approx 0.087$$

conclusion

Very accurate and consistent model

②

Low bias



Low variance

* distance from center shows the bias
* spread of the blue dots show the variance

Case 2): - High bias, low variance

-x-x-

All models predict constant $\hat{f}_i(x) = 65$

predictions at $x=4$

$$\hat{f}_1(4) = \hat{f}_2(4) = \hat{f}_3(4) = 65$$

Mean prediction

$$E[\hat{f}_i(x)] = 65$$

$$\text{Bias}(4) = 65 - 70 = -5; \text{Bias}^2 = 25$$

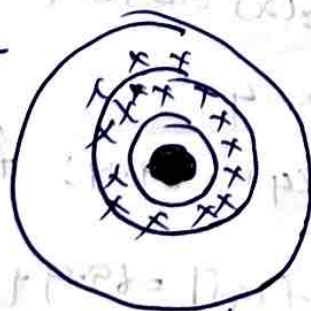
$$\text{Variance} = \frac{1}{3} [(65 - 65)^2 + (65 - 65)^2 + (65 - 65)^2]$$

$$= 0$$

Conclusion: - Model is too simple and fails to learn the true pattern.

Third case

Low bias



High variance

• True predictions
x $\rightarrow$ Actual model prediction

Case 3:- Low bias, High Variance:-

$$\text{Model } \hat{f}_1(4) = 67$$

$$\hat{f}_2(4) = 73$$

$$\hat{f}_3(4) = 71$$

Mean prediction:-

$$E[\hat{f}(4)] = \frac{67+73+71}{3} = 70.33$$

$$\text{Bias}(4) = 70.33 - 70 = 0.33$$

$$\text{Bias}^2 = 0.11$$

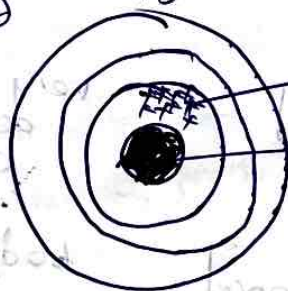
Variance:-

$$\text{Var}(4) = 6.22$$

Conclusion:- Model predictions vary a lot between the datasets, even though the average is close to True value.

Case (2)

High bias



Model prediction
Actual prediction

Case (4):-

$$\text{Model:- } \hat{f}_1(4) = 60$$

$$\hat{f}_2(4) = 15$$

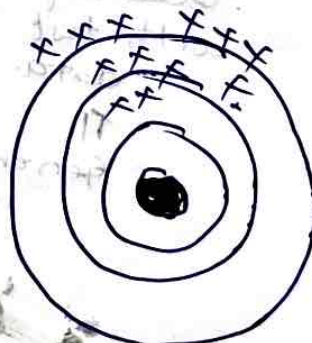
$$\hat{f}_3(4) = 75$$

$$E[\hat{f}(4)] = 51.67$$

$$\text{Bias} = -3.33; \text{Bias}^2 = 11.1$$

$$\text{Var} = 38.89$$

High bias



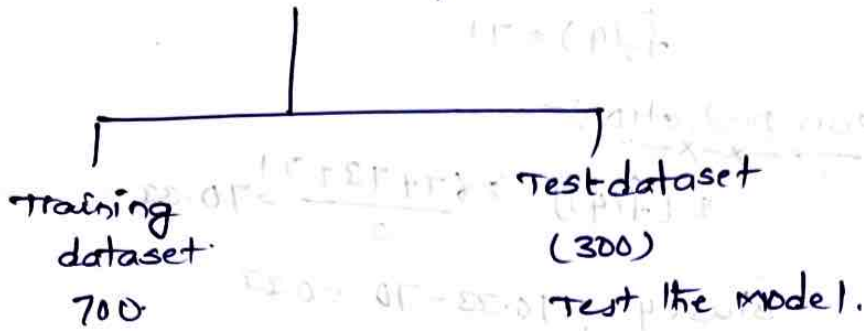
High variance

Conclusion:- Model is both wrong on average and unstable across datasets

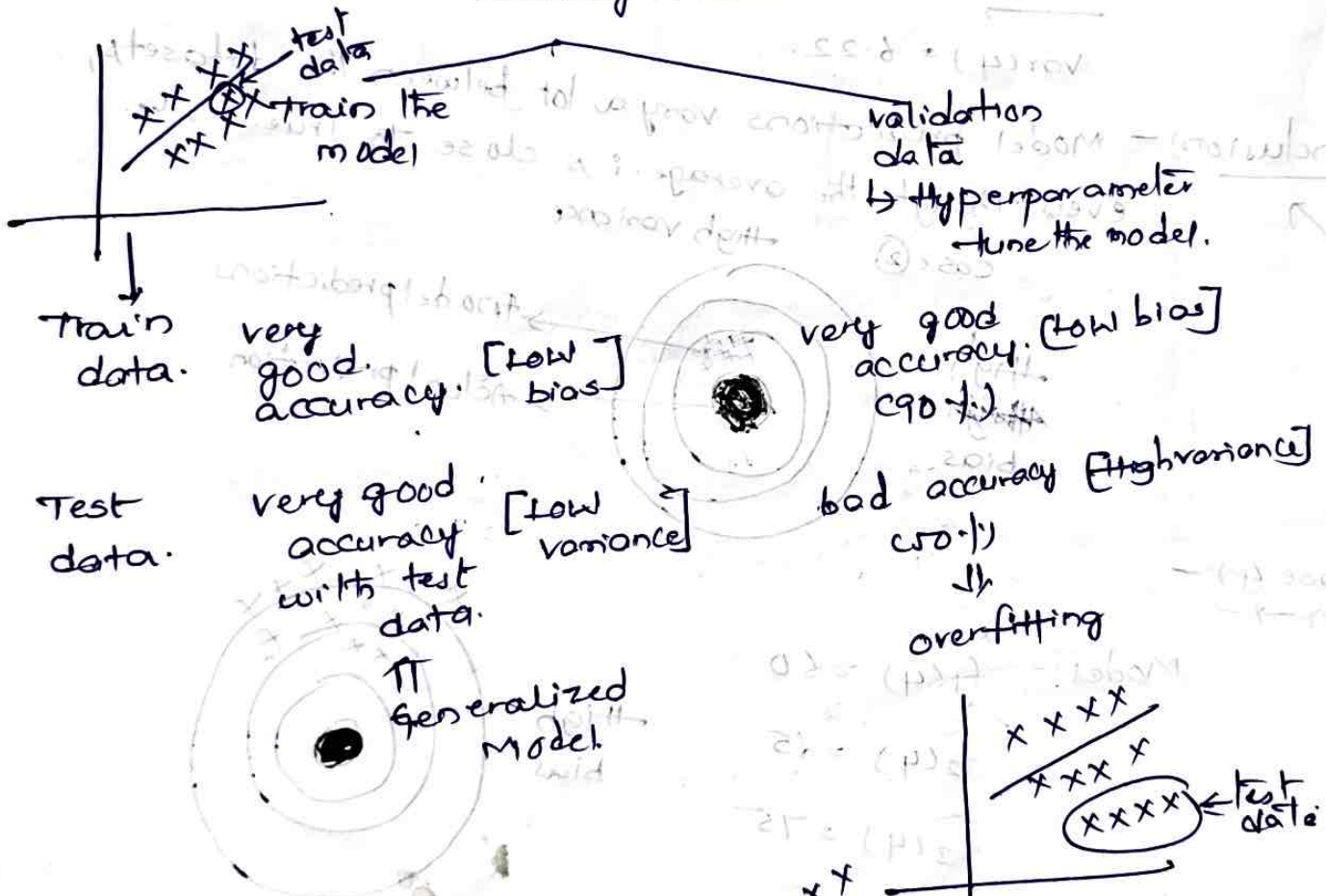
overfitting and underfitting!

Dataset

1000 data points



Training dataset



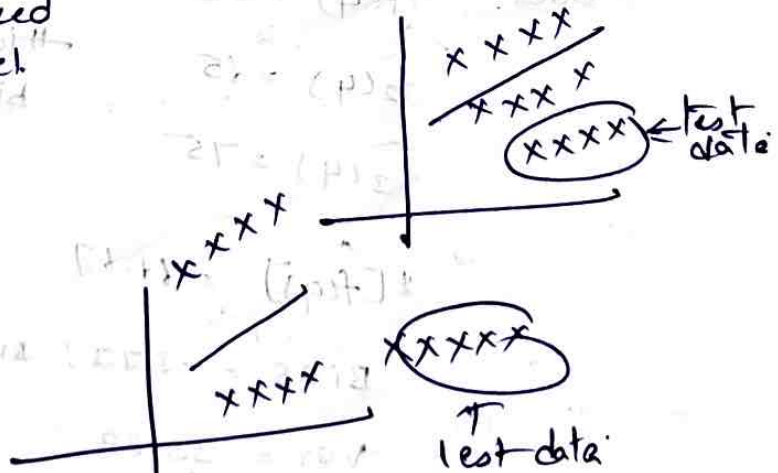
train data model accuracy is low.

test data model accuracy is low

↑

High bias

High variance



problem definition: -

Assume

independent variable = x

dependent variable = y (y depends on ' x ')
 $y = f(x) + \epsilon$

$$y = f(x) + \epsilon$$

Let us assume y value also effected by Noise.

* Noise is modelled by a random variable with zero mean and variance σ^2 .

Magnitude of variance represents the level of uncertainty.

$$\Rightarrow E[\epsilon] = 0$$

$$\text{Var}(\epsilon) = E[(\epsilon - E(\epsilon))^2] = \sigma^2$$

our aim is to find a function $\hat{f}$ as close to f .

$\Rightarrow \hat{f} \leftarrow$ to determine (to find a function $\hat{f}$ such that it is close to the true function f).

$\rightarrow$ function $\hat{f}$ is learnt from the training data by minimizing a loss function.

Goal: - bring predictions of training data as close to their observed value.

$$y = f(\hat{x})$$

$\uparrow$ observed value $\uparrow$ predicted value

Loss-function (mean-squared error)

$$\text{MSE} = E[(y - \hat{f}(x))^2]$$

$$\Rightarrow E[(f(x) + \epsilon - \hat{f}(x))^2]$$

$$\Rightarrow E[(f(x) - \hat{f}(x))^2] + E[\epsilon^2] + 2E[(f(x) - \hat{f}(x))\epsilon]$$

$$\Rightarrow E[(f(x) - \hat{f}(x))^2] + \sigma^2 + 2E[(f(x) - \hat{f}(x))E[\epsilon]]$$

$\uparrow$
0

$$E[(f(x) - \hat{f}(x))^2]$$

$$\Rightarrow E[(f(x) - E[\hat{f}(x)] - (\hat{f}(x) - E[\hat{f}(x)]))^2]$$

subtract and add

$$E[\hat{f}(x)]$$

$$\Rightarrow E[(E[\hat{f}(x)] - f(x))^2] + E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

$$- 2E[(f(x) - E[\hat{f}(x)])(\hat{f}(x) - E[\hat{f}(x)])]$$

0

$$\Rightarrow \text{bias}^2 + \text{var}$$

$$\Rightarrow \text{bias} [f(x)]^2 + \text{var} \hat{f}(x) + e^2$$

inherent noise in the problem

total error

error from wrong model assumptions

error from sensitivity due to fluctuations in training data

Noise coming from data quality

or inaccuracies in data collection

Receiver Operating Characteristic (ROC) Curve: Numerical Example

Problem

Consider the following binary classification problem.

Sample	Actual Class	Predicted Probability	Predicted Score
A	Positive	0.95	
B	Positive	0.90	
C	Negative	0.85	
D	Positive	0.80	
E	Negative	0.70	
F	Positive	0.60	
G	Negative	0.55	
H	Negative	0.40	
I	Positive	0.30	
J	Negative	0.20	

Total Positive samples (P) = 5

Total Negative samples (N) = 5

The ROC curve is obtained by varying the classification threshold.

Performance Measures

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

where

- TP = True Positive
- FP = False Positive
- TN = True Negative
- FN = False Negative

Threshold = 0.90

Predicted Positive: A, B

Confusion Matrix (Predicted along rows and Actual along columns)

Predicted / Actual	Positive	Negative
Positive	2	0
Negative	3	5

Therefore,

$$TP = 2, FP = 0, FN = 3, TN = 5$$

$$TPR = \frac{2}{5} = 0.40$$

$$FPR = \frac{0}{5} = 0$$

ROC Point

$$(0, 0.40)$$

Threshold = 0.80

Predicted Positive: A,B,C,D

Predicted / Actual	Positive	Negative
Positive	3	1
Negative	2	4

$$TP = 3, FP = 1, FN = 2, TN = 4$$

$$TPR = \frac{3}{5} = 0.60$$

$$FPR = \frac{1}{5} = 0.20$$

ROC Point

$$(0.20, 0.60)$$

Threshold = 0.70

Predicted Positive: A,B,C,D,E

Predicted / Actual	Positive	Negative
Positive	3	2
Negative	2	3

$$TP = 3, FP = 2, FN = 2, TN = 3$$

$$TPR = \frac{3}{5} = 0.60$$

$$FPR = \frac{2}{5} = 0.40$$

ROC Point

$$(0.40, 0.60)$$

Threshold = 0.60

Predicted Positive: A,B,C,D,E,F

Predicted / Actual	Positive	Negative
Positive	4	2
Negative	1	3

$$TP = 4, FP = 2, FN = 1, TN = 3$$

$$TPR = \frac{4}{5} = 0.80$$

$$FPR = \frac{2}{5} = 0.40$$

ROC Point

$$(0.40, 0.80)$$

Threshold = 0.55

Predicted Positive: A,B,C,D,E,F,G

Predicted / Actual	Positive	Negative
Positive	4	3
Negative	1	2

$$TPR = \frac{4}{5} = 0.80$$

$$FPR = \frac{3}{5} = 0.60$$

ROC Point

$$(0.60, 0.80)$$

Threshold = 0.40

$$TP = 4, FP = 4, FN = 1, TN = 1$$

$$TPR = 0.80, \quad FPR = 0.80$$

ROC Point

$$(0.80, 0.80)$$

Threshold = 0.30

$$TP = 5, FP = 4, FN = 0, TN = 1$$

$$TPR = 1.0, \quad FPR = 0.80$$

ROC Point

$$(0.80, 1.0)$$

Threshold = 0.20

All samples become Positive.

$$TP = 5, FP = 5, FN = 0, TN = 0$$

$$TPR = 1.0, \quad FPR = 1.0$$

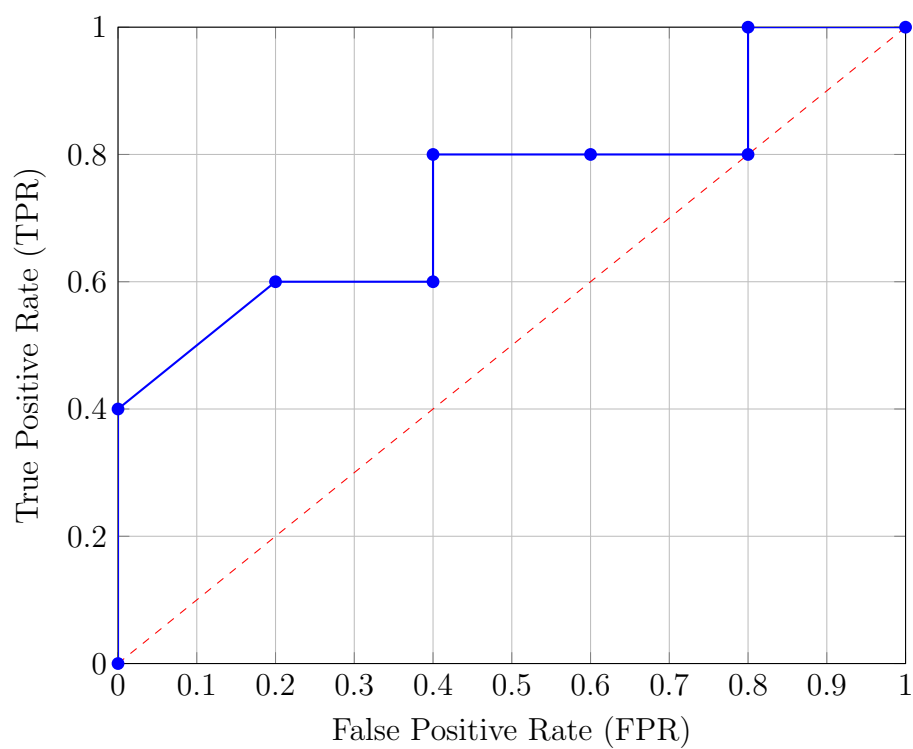
ROC Point

$$(1.0, 1.0)$$

Summary Table

Threshold	TP	FP	FN	TN	TPR	FPR	ROC Point
1.00	0	0	5	5	0.00	0.00	(0.00,0.00)
0.90	2	0	3	5	0.40	0.00	(0.00,0.40)
0.80	3	1	2	4	0.60	0.20	(0.20,0.60)
0.70	3	2	2	3	0.60	0.40	(0.40,0.60)
0.60	4	2	1	3	0.80	0.40	(0.40,0.80)
0.55	4	3	1	2	0.80	0.60	(0.60,0.80)
0.40	4	4	1	1	0.80	0.80	(0.80,0.80)
0.30	5	4	0	1	1.00	0.80	(0.80,1.00)
0.20	5	5	0	0	1.00	1.00	(1.00,1.00)

ROC Curve



Area Under the ROC Curve (AUC)

The Area Under the ROC Curve (AUC) is computed using the **Trapezoidal Rule**. The area between two consecutive ROC points is

$$\text{Area} = \frac{(TPR_1 + TPR_2)}{2} \times (FPR_2 - FPR_1)$$

The ROC points obtained are

(0, 0), (0, 0.40), (0.20, 0.60), (0.40, 0.60), (0.40, 0.80), (0.60, 0.80), (0.80, 0.80), (0.80, 1.00), (1.00, 1.00)

The area between each pair of consecutive ROC points is calculated as follows.

Between Point 1 and Point 2

$$A_1 = \frac{(0 + 0.40)}{2} \times (0 - 0) = 0$$

Between Point 2 and Point 3

$$A_2 = \frac{(0.40 + 0.60)}{2} \times (0.20 - 0) = 0.50 \times 0.20 = 0.10$$

Between Point 3 and Point 4

$$A_3 = \frac{(0.60 + 0.60)}{2} \times (0.40 - 0.20) = 0.60 \times 0.20 = 0.12$$

Between Point 4 and Point 5

$$A_4 = \frac{(0.60 + 0.80)}{2} \times (0.40 - 0.40) = 0$$

Since the width is zero, the area is zero.

Between Point 5 and Point 6

$$A_5 = \frac{(0.80 + 0.80)}{2} \times (0.60 - 0.40) = 0.80 \times 0.20 = 0.16$$

Between Point 6 and Point 7

$$A_6 = \frac{(0.80 + 0.80)}{2} \times (0.80 - 0.60) = 0.80 \times 0.20 = 0.16$$

Between Point 7 and Point 8

$$A_7 = \frac{(0.80 + 1.00)}{2} \times (0.80 - 0.80) = 0$$

Since the width is zero, the area is zero.

Between Point 8 and Point 9

$$A_8 = \frac{(1.00 + 1.00)}{2} \times (1.00 - 0.80) = 1.00 \times 0.20 = 0.20$$

Total Area Under the Curve

Adding all the trapezoidal areas,

$$\begin{aligned} AUC &= A_1 + A_2 + A_3 + A_4 + A_5 + A_6 + A_7 + A_8 \\ &= 0 + 0.10 + 0.12 + 0 + 0.16 + 0.16 + 0 + 0.20 \\ &= 0.74 \end{aligned}$$

Hence,

$$\boxed{AUC = 0.74}$$

Interpretation:

- $AUC = 1.0$: Perfect classifier.
- $AUC = 0.5$: Random classifier.
- $AUC > 0.5$: Better than random guessing.
- In this example, the classifier achieves

$$\boxed{AUC = 0.74,}$$

which indicates **fair to good classification performance**.

1 Selection of the Optimal Threshold

The optimal threshold depends on the application. Different applications have different objectives. Some applications require detecting every positive sample, while others require maintaining a balance between correctly detecting positive samples and minimizing false alarms.

1.1 Scenario 1: Medical Disease Screening

Suppose a hospital develops a machine learning model to detect a serious disease such as cancer.

In this application, failing to detect a patient with the disease (False Negative) is much more dangerous than incorrectly diagnosing a healthy person (False Positive). Therefore, the primary objective is to maximize the True Positive Rate (TPR), even if the False Positive Rate (FPR) increases.

From Table ??,

$$\max(TPR) = 1.00$$

This maximum TPR is achieved at thresholds

$$0.30 \quad \text{and} \quad 0.20.$$

The corresponding performance is

Threshold	TPR	FPR
0.30	1.00	0.80
0.20	1.00	1.00

Although both thresholds correctly detect every patient, threshold 0.30 produces fewer false positives than threshold 0.20. Therefore,

Selected Threshold = 0.30

because all diseased patients are detected while reducing unnecessary false alarms.

Interpretation:

The hospital prefers to perform a few additional medical tests on healthy patients rather than miss a patient who actually has the disease.

1.2 Scenario 2: Balanced Classification (General Machine Learning)

Consider an email spam classifier or a credit approval system.

In these applications, both False Positives and False Negatives are important. Therefore, the objective is to obtain a balanced classifier that simultaneously achieves

- High True Positive Rate
- Low False Positive Rate

A commonly used criterion is the **Youden Index**

$$J = TPR - FPR.$$

The threshold with the largest value of J is selected.

Threshold	TPR	FPR	J
0.90	0.40	0.00	0.40
0.80	0.60	0.20	0.40
0.70	0.60	0.40	0.20
0.60	0.80	0.40	0.40
0.55	0.80	0.60	0.20
0.40	0.80	0.80	0.00
0.30	1.00	0.80	0.20
0.20	1.00	1.00	0.00

The maximum Youden Index is

$$J = 0.40,$$

which occurs at thresholds

$$0.90, 0.80, 0.60.$$

Among these, threshold 0.60 provides the highest sensitivity while maintaining a reasonable False Positive Rate.

Therefore,

Selected Threshold = 0.60

since it offers a good trade-off between detecting positive samples and avoiding excessive false alarms.

Interpretation:

Threshold 0.60 correctly detects 80% of the positive samples while incorrectly classifying only 40% of the negative samples as positive.

Conclusion

The threshold depends on the application.

- **Medical diagnosis or disease screening:**

Choose a threshold that maximizes the True Positive Rate.

Threshold = 0.30

- **General machine learning applications requiring balanced performance:**

Choose the threshold that maximizes the Youden Index.

Threshold = 0.60

2 ROC Curve for Multiclass Classification

The Receiver Operating Characteristic (ROC) curve is naturally defined for binary classification problems. For a multiclass classification problem, the **One-vs-Rest (OvR)** strategy is commonly used to construct ROC curves.

Suppose the dataset contains three classes:

Class A, Class B, Class C

The classifier predicts the probability of each class for every test sample.

Sample	Actual Class	$P(A)$	$P(B)$	$P(C)$
1	A	0.90	0.05	0.05
2	A	0.80	0.10	0.10
3	A	0.40	0.35	0.25
4	B	0.30	0.60	0.10
5	B	0.20	0.70	0.10
6	B	0.40	0.45	0.15
7	C	0.20	0.30	0.50
8	C	0.10	0.20	0.70
9	C	0.30	0.20	0.50

Notice that for every sample,

$$P(A) + P(B) + P(C) = 1.$$

2.1 ROC Curve for Class A

To construct the ROC curve for Class A,

- Positive class = Class A
- Negative class = Class B + Class C

The binary labels become

Sample	Actual Class	Binary Label
1	A	Positive
2	A	Positive
3	A	Positive
4	B	Negative
5	B	Negative
6	B	Negative
7	C	Negative
8	C	Negative
9	C	Negative

Only the probabilities corresponding to Class A are used.

Sample	Binary Label	Probability ($P(A)$)
1	Positive	0.90
2	Positive	0.80
3	Positive	0.40
4	Negative	0.30
5	Negative	0.20
6	Negative	0.40
7	Negative	0.20
8	Negative	0.10
9	Negative	0.30

Using different thresholds, confusion matrices are constructed and the corresponding True Positive Rate (TPR) and False Positive Rate (FPR) are computed to obtain the ROC curve for Class A.

2.2 ROC Curve for Class B

Similarly,

- Positive class = Class B
- Negative class = Class A + Class C

Only the probabilities corresponding to Class B are used.

Sample	Binary Label	Probability ($P(B)$)
1	Negative	0.05
2	Negative	0.10
3	Negative	0.35
4	Positive	0.60
5	Positive	0.70
6	Positive	0.45
7	Negative	0.30
8	Negative	0.20
9	Negative	0.20

The ROC curve is generated by varying the threshold and calculating the corresponding TPR and FPR values.

2.3 ROC Curve for Class C

Finally,

- Positive class = Class C
- Negative class = Class A + Class B

Only the probabilities corresponding to Class C are used.

Sample	Binary Label	Probability ($P(C)$)
1	Negative	0.05
2	Negative	0.10
3	Negative	0.25
4	Negative	0.10
5	Negative	0.10
6	Negative	0.15
7	Positive	0.50
8	Positive	0.70
9	Positive	0.50

Again, different thresholds are selected to calculate the confusion matrices, TPR, and FPR values, resulting in the ROC curve for Class C.

2.4 Final ROC Curves

Using the One-vs-Rest approach, one ROC curve is generated for each class.

Class	Positive vs Negative
Class A	A vs (B,C)
Class B	B vs (A,C)
Class C	C vs (A,B)

Hence, for a classification problem with K classes,

$$\boxed{\text{Number of ROC Curves} = K}$$

Each ROC curve has its own Area Under the Curve (AUC). For example,

Class	AUC
Class A	0.96
Class B	0.91
Class C	0.88

A larger AUC indicates better discrimination capability for that particular class.

Conclusion

The One-vs-Rest (OvR) strategy converts a multiclass classification problem into multiple binary classification problems. Each class is treated as the positive class once, while all remaining classes are grouped as the negative class. Therefore, if there are K classes, K ROC curves are obtained, each with its own AUC value.